

Software Development — Guided Exercise 2

Test Case Documentation

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1. Method 1 — Register Project

This method records projects associated with companies. Validation is performed on all six input parameters. The testing technique applied is Equivalence Class (EC) analysis combined with Boundary Value (BV) analysis.

1.1 Method Signature

```
register_project(company_cif: str, project_acronym: str, project_description: str, department: str, date: str, budget: float)
```

1.2 Equivalence Classes and Boundary Values

The following sections enumerate all valid and invalid equivalence classes and boundary values for each input parameter and the method outputs.

1.2.1 *company_cif*

- ECV1 – String
- ECV2 – Length = 9
- ECV3 – First character is a letter A–Z
- ECV4 – Positions 2–8 are all digits 0–9
- ECV5 – Last character passes the CIF validation algorithm
- ECNV1 – Not a string
- ECNV2 – Length < 9
- ECNV3 – Length > 9
- ECNV4 – First character is not a letter
- ECNV5 – Non-digit found in positions 2–8
- ECNV6 – Last character fails CIF algorithm
- BLNV1 – Length = 8
- BLNV2 – Length = 10

1.2.2 *project_acronym*

- ECV7 – String
- ECV8 – $5 \leq \text{length} \leq 10$
- ECV9 – Characters only uppercase A–Z or digits 0–9
- BLV1 – limit value 5
- BLV2 – limit value 10
- ECNV7 – Not a string
- ECNV8 – Length < 5
- ECNV9 – Length > 10
- ECNV10 – Contains invalid characters (e.g. lowercase, special chars)
- BLNV1 – 4 chars
- BLNV2 – 11 chars

1.2.3 *project_description*

- ECV10 – String

- ECV11 – $10 \leq \text{length} \leq 30$
- BLV3 – limit value 10
- BLV4 – limit value 30
- ECNV11 – Not a string
- ECNV12 – Length < 10
- ECNV13 – Length > 30
- BLNV3 – 9 chars
- BLNV4 – 31 chars

1.2.4 department

Valid classes: ECV12 – String; ECV13 – Value is HR, FINANCE, LEGAL, or LOGISTICS.
Invalid classes: ECNV14 – Not a string; ECNV15 – Any other value.

1.2.5 date

Valid classes: ECV14 – String; ECV15 – Format DD/MM/YYYY; ECV16 – $01 \leq DD \leq 31$; ECV17 – $01 \leq MM \leq 12$; ECV18 – $2025 \leq YYYY \leq 2027$; ECV19 – Date is on or after the request date. Boundary values: BLV6 limit value DD=01, BLV7 limit value DD=31, BLV8 limit value MM=01, BLV9 limit value MM=12, BLV10 limit value YYYY=2025, BLV11 limit value YYYY=2027.

Invalid classes: ECNV16 – Not a string; ECNV17 – Wrong format; ECNV18 – $DD < 01$; ECNV19 – $DD > 31$; ECNV20 – $MM < 01$; ECNV21 – $MM > 12$; ECNV22 – $YYYY < 2025$; ECNV23 – $YYYY > 2027$; ECNV24 – Date before request date. Boundary values: BLNV7 DD=00, BLNV8 DD=32, BLNV9 MM=00, BLNV10 MM=13, BLNV11 YYYY=2024, BLNV12 YYYY=2028.

1.2.6 budget

Valid classes: ECV20 – Float; ECV21 – Exactly 2 decimal places; ECV22 – $50000.00 \leq \text{budget} \leq 1000000.00$. Boundary values: BLV12 limit value 50000.00, BLV13 limit value 1000000.00.

Invalid classes: ECNV25 – Not a float; ECNV26 – Not exactly 2 decimal places; ECNV27 – $\text{budget} < 50000.00$; ECNV28 – $\text{budget} > 1000000.00$. Boundary values: BLNV13 – 49999.99, BLNV14 – 1000000.01.

1.2.7 Output

Valid classes: ECV23 – Returns MD5 string (32 hexadecimal characters) corresponding to input data; ECV24 – Data is stored in corporate_operations.json.

Invalid classes: ECNV29 – EnterpriseManagementException: Invalid Company CIF; ECNV30 – Invalid Project Acronym; ECNV31 – Invalid Project Description; ECNV32 – Invalid Department; ECNV33 – Invalid Date; ECNV34 – Invalid Budget; ECNV35 – Project with same name for same CIF already existed in the data file.

1.3 Test Cases Summary

59 test cases were defined covering all equivalence classes and boundary values. The full test case table is provided in the accompanying Excel file (GE2 2026 F1).

2. Method 2 — Register Document

To be completed for GE3

3. Method 3 — Calculate Project Expenses

To be completed for GE3